SCAMSHIELD AI

SCAMSHIELD MY

Explainable, Community-Verified Scam Intelligence and Intervention Platform

COMPLETE UCRIX INNOVATION PROPOSAL

Analyse. Explain. Verify. Act before the user clicks, shares or pays.

Prepared for UCRIX Innovation 2026

Team: ____________________________ Mentor: ____________________________

Competition proposal draft | Malaysia | 2026

Document purpose and proposal basis

> **Recommended competition identity**
> ScamShield MY is an evolved ScamShield AI proposal that combines explainable scam analysis with ScamRadar's verified community intelligence. The system is positioned as an intervention and digital-safety platform, not merely another scam classifier.

This document develops the attached ScamShield AI and ScamRadar concepts into a complete competition proposal. Source-derived elements include multi-format analysis, explainable risk scoring, Malaysian scam-pattern intelligence, community verification, an elderly-friendly mode, reporting, trend dashboards and safe-action guidance. The detailed architecture, prototype plan, scoring model, budget and implementation roadmap are proposed design assumptions that must be validated through user research and technical testing.

The proposal intentionally avoids claiming that artificial intelligence can guarantee whether a message is safe or fraudulent. ScamShield MY supports human judgement by explaining evidence, recommending verification steps and escalating confirmed cases through a moderated community workflow.

Contents

1. Executive summary and project snapshot

ScamShield MY is an explainable, community-verified scam intelligence and intervention platform designed to help users pause, understand risk and take safer action before clicking a link, scanning a QR code, sharing credentials or transferring money. Users can paste text or submit a screenshot, URL, QR code, phone number or payment request. The system extracts evidence, calculates a hybrid risk score, highlights suspicious indicators in plain language and recommends verification, blocking, reporting or guardian-support actions.

The proposal combines two complementary ideas. ScamShield AI contributes multi-format analysis, explainable prevention and a clear risk dashboard. ScamRadar contributes verified community intelligence, reputation-based moderation, local trend mapping and campus-wide alert potential. Together, they form a more distinctive platform that learns from confirmed local cases instead of relying only on a static scam database.

> **One-sentence pitch**
> ScamShield MY explains and verifies suspicious digital content, then guides users towards safe action before financial or personal information is lost.

Project element	Proposal definition
Project domain	Cybersecurity, digital safety, financial fraud prevention and digital literacy
Primary users	Students, smartphone users, older adults and family caregivers
Institutional users	Universities, banks, telcos, marketplaces, community organisations and public-safety partners
Core innovation	Explainable prevention + multi-format analysis + verified community intelligence + action guidance
Primary SDG	SDG 16 - Peace, Justice and Strong Institutions
Supporting SDGs	SDG 9 - Industry, Innovation and Infrastructure; SDG 10 - Reduced Inequalities; SDG 4 - Quality Education
MVP focus	Text/screenshot analysis, URL/QR checking, risk explanation, action guidance, moderated reporting and trend dashboard
Competition positioning	An explainable digital-safety operating system that helps users stop, verify and act before losses occur

1.1 Why this proposal is competition-ready

- The problem is immediately understandable and emotionally relevant to judges and the public.
- The prototype can demonstrate visible technical work: OCR, QR decoding, URL feature extraction, hybrid scoring, explainability, reporting and analytics.
- The concept has measurable outcomes, including detection performance, user decision improvement, report verification and awareness gains.
- The system can begin as a focused university pilot while retaining a credible path to banks, telcos, marketplaces and community partners.
- The proposal answers the “too common” criticism by shifting from generic detection to explainable intervention and verified local intelligence.

1.2 Recommended project boundaries

> **MVP boundary**
> Build a small but complete end-to-end prototype. Do not attempt automatic access to private messaging applications, real-time bank blocking, deepfake voice detection or nationwide deployment during the competition phase.

2. Background, problem and opportunity

2.1 Problem statement

Scammers impersonate banks, government agencies, employers, delivery services, marketplaces and family members. They often use urgency, fear, attractive rewards, secrecy or authority to pressure users into acting before verification. Students and older adults may become victims before official warnings are circulated, while the same scam pattern can spread rapidly across a campus or local community.

Existing protection is fragmented. A user may need to search a phone number, inspect a domain, read official advisories, ask friends and report the case through separate channels. A simple “scam/not scam” classifier also creates two dangerous outcomes: false positives may cause unnecessary fear, while false negatives may create false confidence.

Problem dimension	Observed gap
Delayed verification	Users act during a high-pressure moment and do not know which official channel to consult.
Fragmented evidence	Text, screenshot, QR destination, URL, phone number and payment details are analysed separately or not at all.
Weak explainability	A probability score alone does not teach users which manipulation techniques or technical indicators created the risk.
Slow local intelligence	New campus or community scam patterns may circulate before static databases are updated.
Under-reporting	Users may not know how to preserve evidence, report safely or warn others without exposing personal data.
Accessibility gap	Older adults and users with low digital confidence need larger text, simpler explanations and optional guardian support.

2.2 Opportunity statement

The opportunity is not to build another isolated detector. It is to create a single, explainable workflow that converts suspicious content into evidence, guidance, community verification and preventive learning. A university pilot is suitable because target users are accessible, controlled testing is possible, campus security or student-affairs units can act as seed verifiers, and before-and-after awareness measurements can be collected.

2.3 Design principles

Principle	Practical meaning
Explain before warning	Every risk result must show the strongest reasons and the supporting evidence.
Pause before action	The interface should interrupt urgency and give the user a clear verification checklist.
Human verification	Community and authorised reviewers confirm reports; AI does not automatically publish accusations.
Privacy by default	Redact personal information, minimise retention and separate public trends from case evidence.
Local learning	Confirmed Malaysian and campus patterns update the knowledge base and educational content.
No false certainty	Use evidence bands and confidence wording; never promise that unflagged content is safe.

3. Proposed solution and innovation

3.1 Solution overview

ScamShield MY provides five connected capabilities: analyse suspicious content, explain the evidence, recommend safe action, verify community reports and convert confirmed cases into local intelligence. The user experience is designed around the moment of uncertainty: “I received this message - what should I notice, how should I verify it and what should I do next?”

Capability	User and system value
Analyse	Accept pasted text, screenshots, URLs, QR codes, phone numbers and payment-request details.
Explain	Highlight urgency, impersonation, credential requests, suspicious domains, payment anomalies and inconsistencies.
Act	Recommend official verification, blocking, evidence preservation, guardian support and appropriate reporting.
Verify	Allow authorised reviewers and reputable community members to confirm, reject, merge or request more evidence.
Learn	Update trend dashboards, educational examples and the local pattern knowledge base using confirmed cases.

3.2 Innovation and differentiation

Differentiator	Why it is stronger than a common scam detector
Explainable prevention	The system does not merely return a label. It reveals which social-engineering and technical indicators increased risk.
Multi-format evidence fusion	Text, OCR output, URL/QR features, payment context, source reputation and community evidence contribute to one assessment.
Community-verified intelligence	Local reports are moderated, deduplicated and confidence-weighted before becoming alerts or trends.
Action-oriented intervention	The result page includes a verification pathway and practical next steps, not only an awareness message.
Inclusive safety mode	A simplified interface uses large text, voice-ready explanations, plain language and optional guardian notifications.
Campus-to-national scalability	The same architecture supports a controlled university pilot and later institutional partnerships without changing the core workflow.

3.3 Product identity and message

> **Recommended tagline**
> Analyse. Explain. Verify. Act before the user clicks, shares or pays.

The project should be presented as a digital-safety intervention platform. Avoid positioning it as an application that “detects every scam”. The memorable competition message is that ScamShield MY helps users make a safer decision at the exact moment manipulation is occurring.

4. Objectives, users and use cases

4.1 Project objectives

1. Develop a functional prototype that analyses at least text, screenshots, URLs and QR codes.
2. Generate an explainable risk assessment that identifies the strongest suspicious indicators in plain language.
3. Recommend safe verification and reporting actions without presenting unverified content as guaranteed safe or fraudulent.
4. Create a moderated community-reporting workflow that increases confidence and produces local scam trends.
5. Design an accessible safety mode for older adults and users with lower digital confidence.
6. Measure technical performance, usability, awareness improvement and decision quality through controlled testing.
7. Demonstrate a credible adoption path for universities and future institutional partners.

4.2 Target users and personas

Persona	Need	Relevant capability
University student	Receives job, parcel, scholarship, rental or marketplace scams; wants a fast explanation before responding.	Mobile scan, URL/QR check, evidence highlighting and reporting.
Older adult	May be targeted by impersonation, investment or family-emergency scams; needs simple language and reassurance.	Large-text safety mode, step-by-step verification and optional guardian alert.
Family caregiver / guardian	Wants to help a family member evaluate high-risk requests without monitoring all private messages.	User-initiated sharing, high-risk notification and action checklist.
Campus security / student affairs	Needs visibility into recurring local scams and a controlled way to issue alerts.	Verified dashboard, duplicate-case merging, trend analytics and broadcast approval.
Moderator / trusted partner	Reviews evidence and confirms or rejects reports.	Case queue, reputation context, audit trail and moderation controls.
Bank, telco or marketplace partner - future	May contribute verified indicators or receive aggregated trends.	Secure institutional integration and anonymised intelligence exchange.

4.3 Priority use cases

ID	Use case
UC-01	Paste a suspicious message and receive highlighted warning signs.
UC-02	Upload a screenshot; OCR extracts text and identifies embedded links or payment details.
UC-03	Scan or upload a QR code and inspect the destination before opening it.
UC-04	Check a URL or phone number against local indicators and technical patterns.
UC-05	Receive a risk band, confidence statement and recommended verification steps.
UC-06	Report a suspicious item with redacted evidence and optional category.
UC-07	Moderator verifies, rejects, merges or requests more evidence.
UC-08	Confirmed cases update campus trend dashboards and learning cards.

ID	Use case
UC-09	Older adult activates simplified safety mode and shares a high-risk case with a guardian.
UC-10	Administrator reviews prototype performance and impact indicators.

5. Functional scope and requirements

5.1 MVP feature set

Priority	Feature	Minimum acceptance condition
MUST	Multi-format submission	Text paste, screenshot upload, URL input and QR image/camera capture.
MUST	Content extraction	OCR, URL extraction, QR decoding, phone-number and payment-request pattern extraction.
MUST	Hybrid risk assessment	Rules, lightweight classifier, local reputation and evidence-confidence fusion.
MUST	Explainable result	Risk band, top indicators, highlighted evidence, uncertainty statement and safe actions.
MUST	Report and moderation	Redacted report, case status, duplicate detection, moderator verification and audit log.
MUST	Trend dashboard	Confirmed case counts, common categories, channels and time-based local trends.
SHOULD	Elderly-friendly mode	Large text, simplified language, clear buttons and optional read-aloud support.
SHOULD	Guardian support	User-initiated sharing or high-risk alert with consent.
SHOULD	Knowledge centre	Short educational cards generated from confirmed patterns.
COULD	Multilingual support	English and Bahasa Malaysia first; Mandarin and Tamil later.
COULD	Institution alert workflow	Authorised campus alert creation after moderator approval.

5.2 Out-of-scope for the competition MVP

- Automatic reading of all WhatsApp, Telegram or encrypted private messages.
- Guaranteed scam detection or automatic financial transaction blocking.
- Deepfake voice and video detection.
- Nationwide law-enforcement case management.
- Publicly displaying personal phone numbers, bank-account information or identifiable victims.
- Unmoderated crowdsourced accusations.
- Full production integration with banks, telcos or government systems.
- Autonomous legal or financial advice.

5.3 Non-functional requirements

Quality attribute	Prototype requirement
Usability	A first-time user completes the primary scan flow without assistance in under two minutes during testing.
Performance	Common text, screenshot and QR analysis returns a prototype result within five seconds on a stable connection.
Security	Encrypted transport, strong authentication for moderators, role-based access and audit logging.
Privacy	Data minimisation, consent, redaction, configurable deletion and no public raw evidence.
Reliability	Clear fallback when OCR, URL lookup or external service is unavailable.
Explainability	Every high-risk result shows at least three evidence-based reasons where available.
Accessibility	Readable contrast, scalable text, plain language, keyboard support on web and touch targets suitable for older adults.
Maintainability	Modular services, versioned rules and documented feature flags.
Ethical safety	No “safe” guarantee; human review required for public alerts and disputed cases.

6. User journeys and system workflows

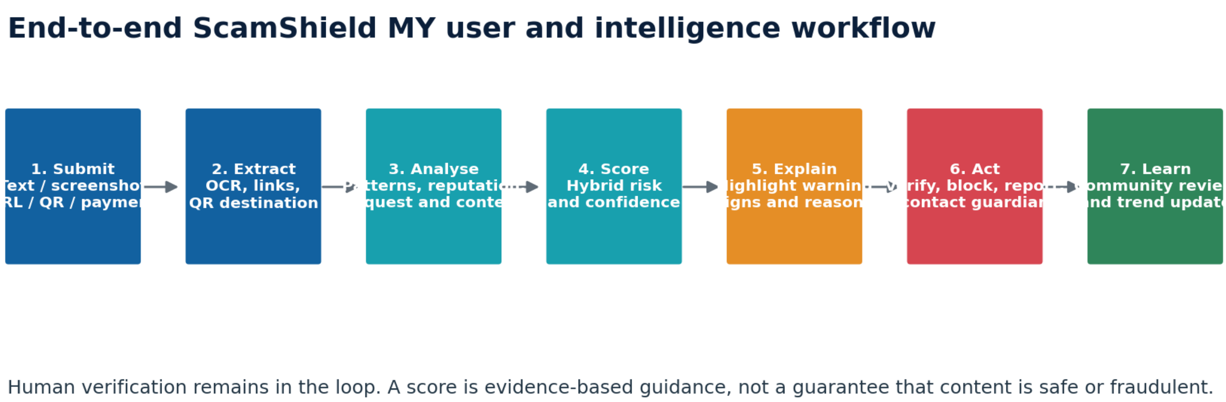

Figure 1. Proposed end-to-end workflow combining ScamShield analysis and ScamRadar verification.

6.1 Primary user journey

1. The user receives suspicious content and opens ScamShield MY voluntarily.
2. The user selects the input type and grants only the permission needed for that action.
3. The system extracts readable text, links, QR destinations, phone numbers and payment-request indicators.
4. The hybrid engine compares the evidence with social-engineering patterns, technical URL features, known local cases and source reputation.
5. The result page displays an evidence band, confidence statement and highlighted warning signs.
6. The safe-action engine recommends official verification, pausing payment, blocking, preserving evidence, guardian support or reporting.
7. If the user reports the case, sensitive fields are redacted and the report enters a moderated queue.

8. Verified cases update local trends and educational content; rejected or disputed cases remain internal for review.

6.2 Community verification workflow

Stage	Control
Submission	User selects category, provides evidence and confirms consent.
Privacy processing	System masks personal details and warns the user before storing sensitive evidence.
Deduplication	Hashing and similarity matching identify repeated URLs, phone numbers, screenshots or message patterns.
Initial status	Case receives “unverified”, “needs evidence” or “high-priority review” status.
Reviewer decision	Authorised reviewer confirms, rejects, merges, escalates or requests more information.
Confidence update	Community or official confirmations increase evidence confidence; disagreements reduce it.
Publication control	Only aggregated, moderated information appears publicly; raw personal evidence remains restricted.
Learning loop	Confirmed patterns are added to the knowledge base after quality review and version control.

7. Technical architecture and technology plan

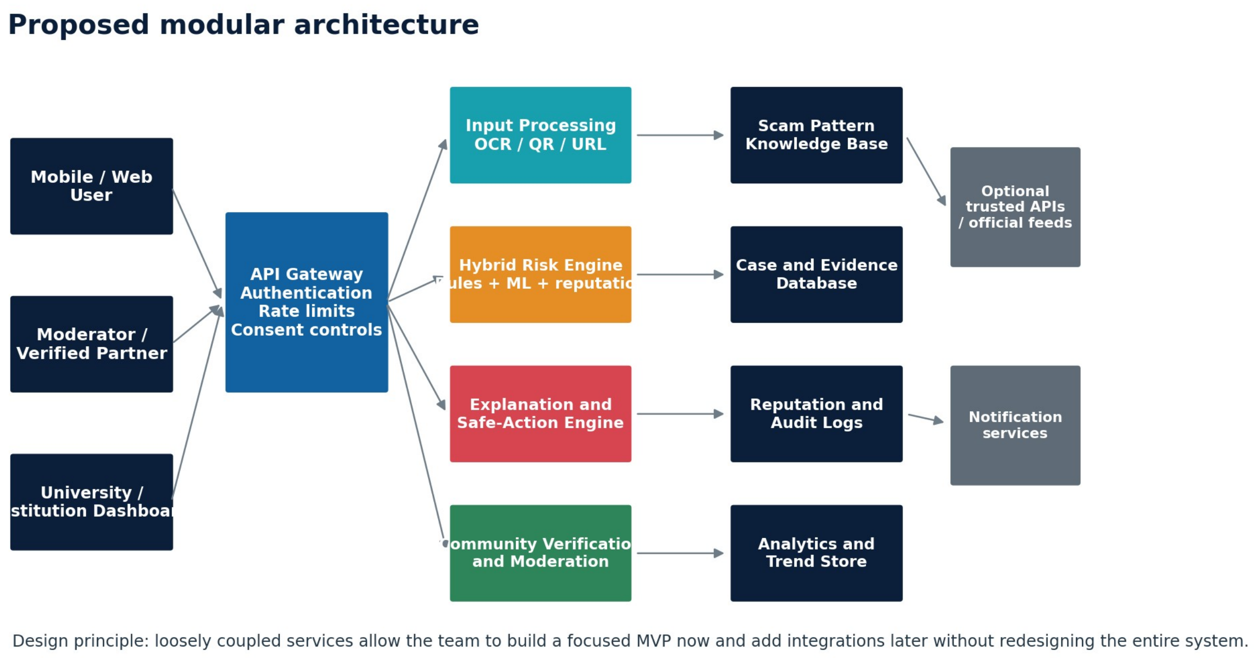

Figure 2. Modular architecture supporting a focused MVP and later institutional integrations.

7.1 Recommended technology stack

Layer	Suggested option	Purpose
Mobile application	Flutter	Single codebase, fast prototyping and strong accessibility support.
Web dashboards	React or Next.js	Suitable for moderator and institutional analytics interfaces.

Layer	Suggested option	Purpose
Backend API	FastAPI or Node.js	Rapid REST API development and straightforward AI-service integration.
Primary database	PostgreSQL	Structured cases, users, evidence metadata, reputation and audit records.
Object storage	Private S3-compatible storage	Restricted screenshots and attachments with expiration policies.
Cache / queue	Redis - optional	Rate limiting, task queues and repeated indicator lookup.
OCR	Tesseract, PaddleOCR or selected cloud OCR	Extract text from screenshots; compare accuracy during prototyping.
QR processing	ZXing or equivalent library	Decode QR contents without opening the destination.
ML / NLP	Python, scikit-learn or lightweight transformer	Classify patterns and support explainable feature contributions.
Explanation layer	Templates plus optional guarded language model	Convert evidence into plain-language explanations; must not override risk rules.
Deployment	Docker on a student-friendly cloud platform	Portable and reproducible competition deployment.
Analytics	Simple SQL dashboard or charting library	Display confirmed trends, categories and prototype KPIs.

7.2 Core data entities

- User and consent record
- Submission and input type
- Extracted indicator
- Risk assessment and model version
- Explanation item
- Recommended action
- Community report
- Evidence attachment
- Moderator decision
- Reputation profile
- Scam pattern
- Notification
- Audit event
- Aggregated trend snapshot

7.3 Integration strategy

The MVP should work without depending on external institutional APIs. A local prototype reputation database and team-curated test cases are sufficient for the live demonstration. Future integrations may provide official advisories, domain reputation, complaint status, telco or marketplace indicators, but every external source must be labelled with provenance and update time.

8. AI, risk scoring and explainability

8.1 Hybrid intelligence approach

A purely generative model should not determine the final risk score. The recommended design combines deterministic rules, a lightweight classifier, technical indicators, source reputation and moderated community evidence. This provides stronger traceability and allows the team to show judges exactly how the result was produced.

Layer	Role in the decision
Rule layer	Detect explicit requests for credentials, OTPs, urgent payment, secrecy, unofficial contact channels, shortened links and common manipulation phrases.
Text classifier	Estimate the probability of scam-related language categories using labelled examples; output remains one input, not the final decision.
Technical feature layer	Inspect URL structure, destination mismatch, suspicious redirection patterns, QR destination and formatting anomalies.
Context layer	Consider claimed sender, requested action, payment method, time pressure and inconsistencies between brand and contact details.
Reputation layer	Check whether indicators match confirmed local cases, official seed data or prior moderated reports.
Community evidence layer	Adjust confidence using verified agreement, evidence quality, duplicate reports and reporter reputation.
Explanation layer	Generate structured reasons from feature contributions and approved templates; optional language generation improves readability only.

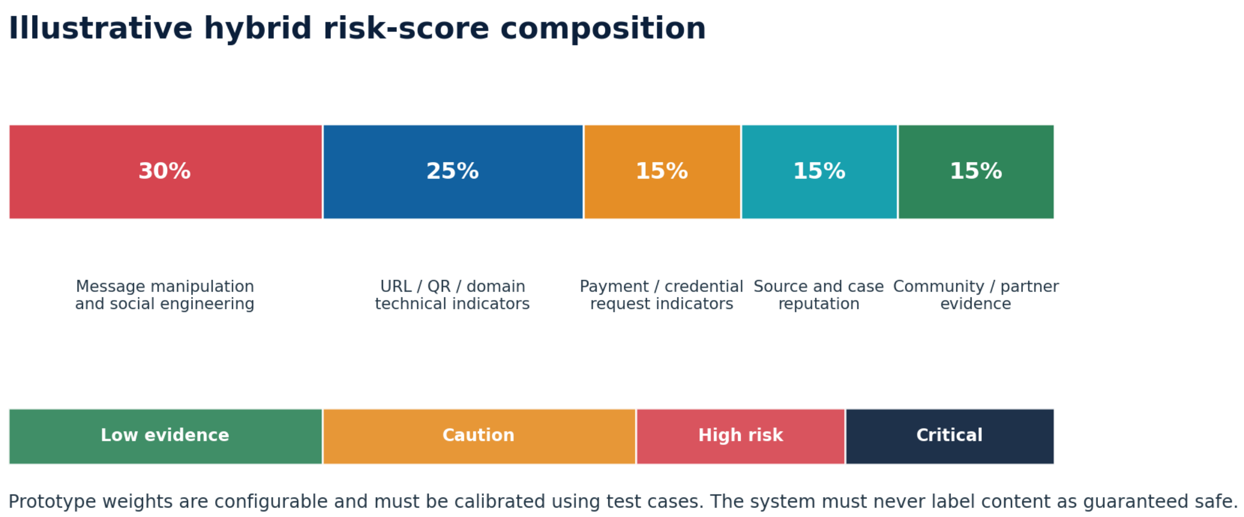

Figure 3. Illustrative prototype risk model. Weights are planning assumptions, not validated production values.

8.2 Risk-band wording

Band	Recommended wording	Safety rule
0-29: Low evidence	No strong indicators detected from the submitted evidence. Verify independently before acting.	Never display “safe”.
30-59: Caution	Suspicious elements were found. Pause and verify using an official channel.	Show missing information and uncertainty.
60-79: High risk	Multiple scam indicators are present. Do not pay or share credentials until independently verified.	Prioritise clear actions and evidence preservation.

Band	Recommended wording	Safety rule
80-100: Critical	Strong combined evidence or verified local matches are present. Block, preserve evidence and report through approved channels.	Escalate to guardian or authorised reviewer with consent.

8.3 Explainability requirements

- Highlight the exact sentence, URL feature, QR destination or payment request that contributed to the score.
- Group reasons into understandable categories: urgency, impersonation, credential request, payment anomaly, technical mismatch and community evidence.
- Show confidence and missing evidence separately from the risk band.
- Provide "Why this matters" explanations suitable for non-technical users.
- Allow users and moderators to inspect the model/rule version used for the result.
- Record whether the explanation changed the user's intended action during testing.

8.4 Data strategy and model development

Data-control area	Recommended practice
Training / validation examples	Use team-curated, synthetic, consented and redacted examples. Avoid storing unnecessary personal data.
Labels	Category, channel, language, manipulation pattern, technical indicator, requested action, verification status and outcome.
Class balance	Include legitimate and ambiguous messages to prevent a model that labels everything suspicious.
Split strategy	Separate training, validation and final test cases; prevent duplicates across splits.
Human review	Two team members review labels; disputed examples receive a documented decision.
Versioning	Store dataset version, feature version, model version and rule version with every assessment.
Bias checks	Compare performance across language, message type, user group and scam category.
Continuous learning	Do not automatically retrain on all reports. Only quality-reviewed, consented and moderated cases enter future datasets.

9. Community verification and intelligence model

9.1 Why community intelligence matters

Scam tactics can change faster than static warning databases. A moderated community layer allows new local patterns to be observed quickly while avoiding the danger of unverified public accusations. The system therefore separates three concepts: AI risk, evidence confidence and publication status.

Concept	Meaning
AI risk	How suspicious the submitted content appears based on current evidence.
Evidence confidence	How complete, consistent and independently supported the evidence is.
Verification status	Unverified, under review, confirmed, rejected, merged, disputed or escalated.

Concept	Meaning
Publication status	Private case, anonymised trend only, campus alert approved or restricted institutional intelligence.

9.2 Reporter and reviewer reputation

Factor	Proposed control
Identity level	Guest, registered user, verified student/staff, approved moderator or institutional partner.
Historical agreement	How often previous reports matched later moderator decisions.
Evidence quality	Completeness, clarity, original source, redaction quality and reproducibility.
Behaviour controls	Rate limits, duplicate detection, abuse reports and suspension history.
Conflict management	Users cannot verify their own reports; sensitive allegations require authorised review.
Decay and appeals	Old reputation gradually decays; users can appeal decisions through a logged process.

9.3 Cold-start strategy

- Seed the prototype with reviewed synthetic and redacted examples representing several common scam categories.
- Invite a small group of verified student testers and one university unit to act as initial reviewers.
- Keep early trend data labelled as demonstration or pilot data.
- Require stronger evidence before a pattern appears as a public alert.
- Use the knowledge centre to provide value even before a large report database exists.

10. User experience, accessibility and compatibility

10.1 Primary interface screens

Screen	Core purpose
Home / quick check	Large input choices: paste text, upload screenshot, scan QR, check URL or report.
Analysis progress	Show extraction steps and permission use; never create the impression that private messages are read automatically.
Result dashboard	Risk band, confidence, highlighted reasons, recommended actions and "verify independently" reminder.
Evidence details	Technical information expandable for advanced users without overwhelming others.
Report flow	Category, redaction preview, consent and case status tracking.
Community trends	Anonymised charts, common scam categories and approved local alerts.
Knowledge centre	Short lessons, examples and "spot the warning signs" interactions.
Moderator dashboard	Queue, duplicate cases, evidence viewer, decision controls, audit log and alert approval.

Screen	Core purpose
Elderly-friendly mode	Large text, simple vocabulary, fewer choices per screen, strong contrast and optional read-aloud output.

10.2 Compatibility plan

Environment	Compatibility approach
Android	Primary mobile target because it supports broad user access and rapid camera/QR prototyping.
iOS	Flutter-based build can be extended; competition testing may prioritise available devices.
Web	Responsive user checker plus full moderator/institution dashboard.
Low-end devices	Compress images, limit background processing and provide manual text input fallback.
Weak connectivity	Allow local QR decoding and queued report submission; clearly mark unavailable reputation checks.
Languages	MVP: English and Bahasa Malaysia interface; later: Mandarin and Tamil.
Assistive technology	Semantic labels, keyboard navigation, scalable text and screen-reader-compatible result structure.

10.3 Behavioural design

ScamShield MY should reduce panic instead of increasing it. Red is reserved for high-confidence risk and urgent action. The interface should first instruct the user to pause, then explain the evidence, and finally offer a small number of safe actions. The system should never shame victims or suggest that a person was careless.

11. Privacy, cybersecurity, ethics and safety boundaries

11.1 Privacy-by-design controls

- Collect only the content required for the selected scan or report.
- Provide a redaction preview before a screenshot or report is stored.
- Mask personal phone numbers, names, account numbers and unrelated chat content.
- Allow immediate analysis without creating a public report.
- Use private object storage, short-lived access links and configurable retention periods.
- Separate identifiable evidence from anonymised trend data.
- Require explicit consent for guardian sharing and institutional escalation.
- Provide deletion and withdrawal controls where operationally possible.
- Log moderator access to sensitive evidence.

11.2 Threat model and controls

Threat	Mitigation
Malicious report poisoning	Verified roles, reputation weighting, rate limits, duplicate detection, moderator review and appeals.
Privacy leakage	Automated and manual redaction, restricted storage, role-based access and no public raw evidence.
Adversarial text or images	Input validation, model confidence thresholds, rule fallbacks and test cases with obfuscation.

Threat	Mitigation
Dangerous links	Never open URLs automatically; analyse structure in a sandboxed or metadata-only process.
Account takeover	Strong authentication, secure password handling, optional multi-factor authentication for moderators.
Model overconfidence	Evidence bands, uncertainty wording, legitimate/ambiguous test cases and no “safe” guarantee.
Biased performance	Language and user-group testing, documented limitations and human escalation.
Abusive public accusations	Private moderation, aggregation, proof thresholds and restricted publication.
Data breach	Encryption in transit, secrets management, least privilege, backups and incident-response plan.

11.3 Ethical boundaries and disclaimers

> **Required user-facing boundary**
> ScamShield MY provides a preliminary, evidence-based risk assessment and digital-safety guidance. It cannot guarantee that content is legitimate or fraudulent and does not replace verification through official organisations.

- AI results must not trigger public accusations automatically.
- The system must not expose victims or suspected individuals through public raw data.
- High-stakes decisions should be referred to official banks, telcos, platforms or authorities.
- Users must be encouraged to contact trusted official channels rather than phone numbers contained in suspicious messages.
- The team must document limitations honestly during judging.

12. Prototype scope and live demonstration

12.1 Recommended competition MVP

The strongest UCRIX prototype is a focused end-to-end system rather than a collection of disconnected mock-ups. The MVP should support one user application and one moderator dashboard, with a curated set of legitimate, scam and ambiguous test cases.

Prototype component	Deliverable
User application	Text paste, screenshot upload, QR scan/upload, URL input, risk result, reasons, actions and report button.
Analysis services	OCR, QR decoding, URL parsing, pattern rules, lightweight classifier and configurable risk fusion.
Community module	Report submission, redaction preview, duplicate indicator matching and case-status tracking.
Moderator dashboard	Case queue, evidence review, confirm/reject/merge decision and alert approval.
Trend dashboard	Confirmed case counts, categories, channels and a simple campus trend map or faculty-level display.
Accessibility	Elderly-friendly display mode and plain-language result explanation.
Testing evidence	Accuracy/confusion matrix, usability task results, pre/post awareness survey and user feedback quotes.

Three-minute live demonstration storyline

1 **0:00-0:30 Show the problem**
A student receives a delivery-payment screenshot with an urgent QR code.

2 **0:30-1:10 Analyse**
Upload screenshot; OCR extracts text and decodes the QR destination.

3 **1:10-1:45 Explain**
Risk score appears with highlighted urgency, unofficial domain and payment request.

4 **1:45-2:20 Act**
System recommends official verification, evidence preservation and guardian/report options.

5 **2:20-3:00 Verify and learn**
Moderator confirms a duplicate report; campus trend dashboard updates immediately.

Figure 4. Suggested live demonstration sequence designed for immediate judge understanding.

12.2 Demonstration test cases

Scenario	Input	Expected behaviour
Legitimate message	Official-looking but valid message with no payment request.	System shows low evidence, explains limitations and recommends independent verification.
Delivery-payment scam	Urgent screenshot with a QR code and unofficial payment destination.	OCR + QR analysis highlights urgency, destination mismatch and payment pressure.
Job scam	High-salary offer requesting a processing fee and personal documents.	System highlights reward bait, upfront fee and credential/document request.
Family emergency impersonation	Unknown number requests immediate transfer and secrecy.	System identifies emotional pressure, secrecy and payment urgency; offers guardian/family verification steps.
Ambiguous marketplace message	Unfamiliar seller requests off-platform communication.	System returns caution rather than automatic scam classification.
Duplicate community report	Second user reports the same URL or number.	System links the case, raises evidence confidence and updates the moderator dashboard.

13. Testing, validation and measurable impact

13.1 Validation strategy

Validation area	Method
Problem validation	Interview students, older adults/caregivers and one institutional stakeholder about verification difficulties and trusted actions.
Prototype usability	Observe users completing scan, explanation and reporting tasks without coaching.
Technical evaluation	Measure OCR success, QR decoding, URL extraction, classification performance, latency and system errors.

Validation area	Method
Explanation quality	Ask users to identify why the content is risky before and after viewing ScamShield explanations.
Decision impact	Record whether users would click/pay/share before and after the result.
Community workflow	Test duplicate matching, reviewer agreement, abuse handling and publication controls.
Accessibility	Test text size, contrast, comprehension and task completion with older or lower-confidence users.
Security review	Perform role-access tests, file-upload validation, rate-limit checks and privacy/redaction review.

13.2 Suggested KPIs and prototype targets

Impact area	Measurement	Prototype target
High-risk identification	Recall on labelled high-risk test cases	At least 85% in the controlled test set
False-warning control	Specificity on legitimate test cases	At least 80% in the controlled test set
Explanation comprehension	Users correctly name key warning signs after analysis	At least 80%
Decision improvement	Users changing from unsafe to safer intended action	Meaningful pre/post improvement; target 30 percentage points
Usability	Primary scan completed without assistance	At least 85% of test participants
Speed	Median analysis response time	Below five seconds for common prototype cases
Report quality	Submitted reports containing sufficient evidence	At least 75%
Moderator consistency	Agreement between two reviewers on test cases	At least 80%
Community verification	Confirmed duplicate cases linked correctly	At least 90% in prepared scenarios
Awareness	Pre/post scam-awareness assessment	At least 20% average improvement
Privacy	Sensitive fields successfully redacted in prepared cases	At least 95%
System reliability	Successful completion of demo test runs	At least 95%

All numerical targets above are internal prototype goals, not official UCRIX requirements. They should be adjusted after baseline testing and reported honestly with sample size and limitations.

13.3 User-research sample

- 5-10 university students representing different faculties and levels of digital confidence.
- 3-5 older adults or family caregivers, where access and consent are practical.
- 1-2 university staff members from security, student affairs, ICT or counselling-related support.
- At least two team members acting independently as prototype moderators.
- Optional feedback from a cybersecurity lecturer or industry mentor.

14. UCRIX evaluation alignment

The attached project-selection materials emphasise relevance, innovation and creativity, technical execution, measurable impact, feasibility, presentation quality, prototype strength, scalability and judge appeal. ScamShield MY should provide visible evidence for each criterion rather than relying on claims.

Evaluation metric	How ScamShield MY fulfils it	Evidence to present
Problem relevance	Digital scams affect students, families and institutions; the problem is immediately understandable.	User interviews, local examples and a clear before-loss scenario.
Innovation / creativity	Explainable prevention, evidence fusion, community verification and action guidance go beyond a common classifier.	Comparison table against a basic scam checker and a live community-confidence update.
Technical execution	OCR, QR decoding, URL analysis, ML/rules, dashboards, access controls and analytics form a complete system.	Working end-to-end prototype and architecture explanation.
Measurable impact	Safety decisions, awareness, analysis performance, report quality and moderation can be measured.	Pre/post test, confusion matrix, task completion and KPI dashboard.
Feasibility	The MVP uses accessible software tools and a controlled university pilot; expensive integrations are deferred.	Clear scope, timeline, budget and risk controls.
Prototype strength	Judges can submit prepared examples and see analysis, explanation, action and community verification.	Three-minute live demo with fallback recording.
Scalability	Modular architecture supports universities first, then banks, telcos, marketplaces and community organisations.	Phased adoption roadmap and partner roles.
Presentation quality	The narrative is simple: suspicious content -> explanation -> safe action -> verified local learning.	Consistent visual identity, concise pitch and evidence-led storytelling.
Judge appeal	Immediate emotional relevance is combined with responsible AI boundaries and visible technology.	Show both a near-loss prevention story and honest limitations.

14.1 Internal readiness assessment

Area	Internal score / 10	Reason
Relevance	9.5	Strong and immediately understood problem
Innovation	9.0	Strong after combining explainability, verification and intervention
Technical execution	8.5	Multiple working modules; manageable with strict scope
Impact	9.0	Direct prevention and digital-literacy value
Feasibility	8.0	High for software MVP; data quality requires care
Demo quality	9.5	Clear visual before/after flow
Scalability	9.0	Campus pilot to institutional platform
Responsible design	9.0	Human review, privacy controls and uncertainty wording

This is an internal planning assessment, not an official score or prediction of competition results.

15. Feasibility, team structure, timeline and budget

15.1 Student-team feasibility

The complete production vision is large, but the recommended MVP is achievable when the team limits input types, uses a curated dataset, postpones deepfake analysis and treats institutional integrations as future work. The project is especially compatible with computer science, networking, cybersecurity, AI, data and UI/UX skills.

Role	Primary responsibilities
Project lead / product manager	Scope, stakeholder interviews, requirements, sprint coordination, proposal and pitch integration.
AI / data engineer	Dataset preparation, rules, classifier, scoring, explanation evidence and model evaluation.
Backend / security engineer	APIs, database, authentication, storage, logging, moderation and privacy controls.
Frontend / UX engineer	Mobile/web interface, elderly-friendly mode, user journeys and accessibility testing.
Full-stack / research and presentation lead	Dashboard integration, analytics, user testing, documentation, video and live-demo preparation.
Mentor	Technical review, ethics, research guidance, competition coaching and external connection support.

15.2 Ten-week development roadmap

Period	Focus	Key output
Week 1	Problem validation	Interview users, collect scam categories, define success measures and finalise MVP.
Week 2	Requirements and UX	User stories, wireframes, data model, privacy flow and test plan.
Week 3	Foundation	Repository, authentication, database, submission API and basic mobile/web shell.
Week 4	Input processing	OCR, URL extraction, QR decoding, file validation and test fixtures.
Week 5	Risk engine	Rules, lightweight classifier, score fusion, evidence logging and result bands.
Week 6	Explanation and actions	Highlighted indicators, plain-language reasons, verification checklist and report flow.
Week 7	Community module	Moderator queue, duplicate matching, reputation fields and trend dashboard.
Week 8	Integration and accessibility	End-to-end flow, elderly-friendly mode, multilingual interface labels and performance fixes.
Week 9	Testing and refinement	Technical evaluation, user testing, privacy/security review and bug fixing.
Week 10	Competition delivery	Demo data, backup video, pitch deck, proposal refinement and rehearsal.

15.3 Estimated student prototype budget

Item	Estimated RM	Note
Cloud hosting and database	200	Short-term prototype deployment and backups.
Optional OCR / API testing credits	150	Only if open-source processing is insufficient.
Domain and basic deployment services	100	Competition-ready access and HTTPS.
User-testing materials and transport	150	Participant refreshments, printing or travel where appropriate.
Pitch materials and demonstration setup	150	Poster, QR cards, backup hotspot or display accessories.
Contingency	250	Unexpected service, storage or presentation needs.
Estimated total	1,000	Planning estimate; can be reduced through university/free-tier resources.

The budget is an internal planning estimate. It excludes team labour, production-scale security audits and institutional integration costs.

16. Risks, limitations, scalability and adoption

16.1 Risk register

ID	Risk	Impact	Likelihood	Mitigation
R1	False positive	High	Medium	Evidence bands, legitimate test cases, confidence wording and independent verification guidance.
R2	False negative	High	Medium	Never guarantee safety; show limitations and encourage official verification.
R3	Insufficient or biased data	High	Medium	Curated balanced cases, versioning, language testing and human review.
R4	Fake community reports	High	Medium	Reputation, rate limits, duplicate checks, moderator approval and appeals.
R5	Privacy exposure	High	Medium	Redaction, restricted access, minimal retention and no public raw evidence.
R6	Over-scoped prototype	High	High	Strict MUST/SHOULD/COULD priorities and freeze non-MVP features.
R7	External API unavailable	Medium	Medium	Local prototype data and graceful fallback messaging.
R8	OCR or QR failure	Medium	Medium	Manual text/URL fallback and prepared demo devices/cases.

ID	Risk	Impact	Likelihood	Mitigation
R9	Judges view topic as common	High	Medium	Lead with intervention, explainability and community verification; demonstrate differentiation visibly.
R10	Live-demo failure	High	Low-Medium	Offline fallback data, recorded backup and rehearsed reset procedure.

16.2 Practical limitations

- Scam tactics and language change continuously, so the knowledge base and models require regular updates.
- The MVP cannot automatically inspect every private messaging platform due to permissions and encryption.
- Risk results depend on the quality and completeness of submitted evidence.
- Community reports can be wrong, malicious or duplicated without moderation.
- Technical URL reputation may be unavailable for new or short-lived domains.
- Deepfake voice and video scams remain outside the first version.
- Real deployment requires legal, privacy, cybersecurity and institutional governance review.

16.3 Adoption and commercialisation pathway

Phase	Scope	Possible model
Phase 1 - University pilot	Student mobile/web checker, verified campus moderation and local trend dashboard.	Evidence of user value, performance and adoption.
Phase 2 - Multi-campus service	Shared knowledge base, institution-specific alerts and central quality control.	Institutional subscription or sponsored digital-safety programme.
Phase 3 - Partner integrations	Banks, telcos and marketplaces contribute verified indicators and receive aggregated intelligence.	API licensing, enterprise dashboard and partnership agreements.
Phase 4 - Public platform	Broader multilingual access, guardian networks, official reporting links and advanced threat modules.	Public-private collaboration and regional scalability.

16.4 Potential stakeholders and partners

- University ICT, security, student-affairs and residential-college units
- Cybersecurity lecturers and student clubs
- Banks and payment providers
- Telecommunications companies
- Online marketplaces and delivery platforms
- Consumer-protection or digital-literacy organisations
- Community centres and elderly-support groups
- Relevant public authorities for verified reporting guidance

17. SDG alignment and conclusion

17.1 Sustainable Development Goal alignment

SDG	Contribution	Role
SDG 16 - Peace, Justice and Strong Institutions	Reduces vulnerability to fraud, supports responsible reporting and strengthens trust in digital interactions.	Primary

SDG	Contribution	Role
SDG 9 - Industry, Innovation and Infrastructure	Applies AI, cybersecurity and data infrastructure to a high-impact digital-safety problem.	Supporting
SDG 10 - Reduced Inequalities	Provides inclusive protection for older adults and users with lower digital confidence.	Supporting
SDG 4 - Quality Education	Transforms every scan into practical digital-literacy education through explainable warning signs.	Supporting

17.2 Final proposal conclusion

ScamShield MY is strongest when it is not presented as a perfect scam detector. Its competition value comes from integrating explainable AI, technical evidence, community verification, accessible guidance and a measurable learning loop. The proposal is ambitious enough to demonstrate meaningful computer-science innovation but can still be reduced to a realistic university MVP.

The recommended prototype tells a clear story: a user receives suspicious content, the system extracts and explains the evidence, the user chooses a safer action, a moderated report strengthens local intelligence, and the university learns which scam patterns are spreading. This complete prevention workflow provides stronger judge appeal than a standalone classifier and supports credible future adoption.

> **Recommended competition closing line**
> ScamShield MY does not ask users to trust a black-box label. It shows the evidence, slows down manipulation and connects one person's warning to a safer community.

Appendix A. Detailed feature backlog

Module	Feature	Priority	Purpose
Input	Text paste	MVP	Direct message analysis.
Input	Screenshot OCR	MVP	Extracts text and visible indicators.
Input	QR scanner	MVP	Decodes without opening destination.
Input	URL checker	MVP	Parses domain and suspicious structure.
Input	Phone-number checker	Should	Matches moderated local reports.
Input	Payment-request form	Should	Captures account/payment context without storing unnecessary data.
Analysis	Rule engine	MVP	Transparent high-signal patterns.
Analysis	Text classifier	MVP	Supports category and language pattern recognition.
Analysis	URL feature engine	MVP	Technical indicators and destination mismatch.
Analysis	Risk fusion	MVP	Combines evidence with configurable weights.
Analysis	Confidence display	MVP	Separates evidence completeness from risk.
Analysis	Brand-impersonation visual model	Future	Requires stronger image data and validation.

Module	Feature	Priority	Purpose
Analysis	Voice/deepfake detection	Future	Explicitly outside competition scope.
User safety	Evidence highlighting	MVP	Explains exact suspicious elements.
User safety	Action checklist	MVP	Official verification, block, preserve evidence and report.
User safety	Guardian share	Should	Consent-based support for high-risk cases.
User safety	Elderly mode	Should	Large text, plain language and simplified navigation.
User safety	Read-aloud explanation	Could	Accessibility enhancement.
Community	Report submission	MVP	Redacted, consented case submission.
Community	Duplicate detection	MVP	Groups repeated indicators.
Community	Moderator queue	MVP	Human review and status management.
Community	Reporter reputation	Should	Confidence weighting and abuse control.
Community	Campus alert workflow	Could	Authorised publication only.
Community	Public scam map	Future	Use aggregated non-identifying data only.
Education	Knowledge cards	Should	Short explanations from confirmed patterns.
Education	Interactive quiz	Could	Measures awareness improvement.
Institution	Trend dashboard	MVP	Categories, channels and confirmed case trends.
Institution	Partner API	Future	Controlled institutional data exchange.

Appendix B. Proposed data model

Entity	Illustrative fields
users	user_id, role, verification_level, language, accessibility_mode, consent_version, created_at
submissions	submission_id, user_id, input_type, created_at, storage_reference, retention_expiry
indicators	indicator_id, submission_id, type, normalised_value_hash, displayed_redacted_value, extraction_confidence
assessments	assessment_id, submission_id, risk_score, risk_band, confidence, model_version, rule_version, created_at
explanations	explanation_id, assessment_id, category, evidence_reference, contribution, plain_language_text
recommended_actions	action_id, assessment_id, action_type, priority, official_channel_note
reports	report_id, submission_id, category, status, consent, privacy_review_status, created_at
moderation_decisions	decision_id, report_id, reviewer_id, decision, rationale, timestamp

Entity	Illustrative fields
reputation_profiles	profile_id, user_id, verified_status, agreement_rate, evidence_quality, abuse_flags
patterns	pattern_id, category, language, indicators, provenance, verification_status, active_version
notifications	notification_id, recipient_id, type, consent_basis, sent_at, delivery_status
audit_events	event_id, actor_id, action, object_type, object_id, timestamp, security_context
trend_snapshots	snapshot_id, period, location_scope, category_counts, publication_status

Appendix C. Sample explainable analysis output

> **Example input**
> "Your parcel is being held. Pay RM2.50 within 30 minutes using the QR code below or the parcel will be returned." The screenshot displays a delivery-company logo, but the QR destination uses an unrelated shortened domain.

Output component	Displayed result	Explanation
Risk band	High risk - 76/100	Multiple independent indicators are present; this remains a preliminary assessment.
Urgency pressure	High contribution	The message creates a 30-minute deadline to reduce the user's opportunity to verify.
Payment request	Medium contribution	An unexpected payment is requested through a message rather than an official account or application.
QR destination mismatch	High contribution	The decoded destination does not match the claimed delivery-company domain.
Impersonation	Medium contribution	A familiar logo is used, but contact and destination information are inconsistent.
Community evidence	Low/medium contribution	Two similar reports are under review; no public confirmation is shown until moderation is complete.
Recommended action	Do not pay or open the destination	Open the official delivery application manually, check tracking status and contact support through its official channel.
Reporting action	Preserve and report	Save the screenshot, redact personal information and submit it for moderated verification.

Appendix D. User interview guide

1. Tell us about the last suspicious message, call, link or payment request you received.
2. What did you do first, and which part of the decision was most difficult?
3. Which sources or people do you trust when verifying suspicious content?
4. Would you be comfortable uploading a screenshot? Which privacy controls would you need?
5. Would a risk score alone help you, or would you need the reasons and recommended actions?
6. How would you prefer the system to explain uncertainty?
7. Would you report a suspected scam? What would make reporting easier or safer?

8. For older users: which text size, language, audio or guardian-support features are useful?
9. For institutional users: which trend information is useful without exposing individuals?
10. What would make you continue using the platform after the first check?

Appendix E. Prototype test-case matrix

ID	Case	Key condition	Expected result
T01	Legitimate official notification	No payment / credential request	Low evidence; independent verification reminder
T02	Urgent parcel-payment screenshot	QR + urgency + unofficial domain	High risk; QR mismatch highlighted
T03	Job offer with processing fee	Reward + upfront payment	High risk; fee and unrealistic reward explained
T04	Family-emergency message	Unknown number + secrecy + transfer	High risk; family verification action shown
T05	Marketplace off-platform request	Unfamiliar seller + off-platform contact	Caution; not automatically classified as scam
T06	Obfuscated URL	Spacing or punctuation hides destination	URL extraction or manual fallback requested
T07	Low-quality screenshot	OCR confidence is weak	System asks user to crop or paste text
T08	Repeated confirmed URL	Matches moderated pattern	Community evidence contribution shown
T09	Malicious false report	Insufficient evidence / abusive frequency	Rate limit and moderator rejection
T10	Personal data in screenshot	Phone/account details visible	Redaction preview before report storage
T11	No internet	QR decoded locally; reputation unavailable	Partial result with clear limitation
T12	Moderator disagreement	Two different decisions	Case becomes disputed and requires senior review

References and source basis

1. 1. UCRIX Innovation Proposal Pack - ScamShield AI: problem statement, multi-format analysis, explainable prevention, prototype features, limitations, judge appeal and impact indicators.
2. 2. UCRIX 2026 Detailed Brainstorm Summary - ScamRadar: verified community intelligence, local scam map, reputation system, guardian notification and campus-alert workflow.
3. 3. UCRIX / URIIS Project Idea Proposal Pack: project-selection framework covering problem importance, target-user access, prototype feasibility, data availability, technical differentiation, SDG relevance, measurable impact, adoption model, demonstration quality and team capability.
4. 4. UCRIX Three Original Project Proposals: internal evaluation framing using relevance, originality, technical execution, measurable impact, feasibility and presentation quality.

Note: The detailed architecture, weights, target values, timeline, budget and governance controls in this document are proposed planning content developed from the source concepts. They require validation before real-world deployment.